Reclaim

Predicting return risk in e-commerce

Deliverable 1: Problem Definition & Data Foundations

Track A: Application Development 24 October 2026

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Repository: https://github.com/Soban06/Reclaim

1. Problem definition

Online retailers must handle uncertainty about which products will be returned. Returns create reverse-logistics, handling and inventory-planning work. Reclaim proposes a decision-support application that estimates return risk from product and customer context. The topic has been approved for Track A.

The intended users are e-commerce operations managers, inventory planners and customer experience staff. A future interface will accept purchase details, show a risk score and explain model signals. Scores should support planning and assistance rather than automatically refusing orders or return rights.

Prediction target and scope

The initial dataset supports **purchase-event / product-variant return prediction**: $y = 1$ means returned and $y = 0$ means kept. It lacks order IDs connecting products into baskets. This phase therefore does not claim to predict the return of an entire multi-item order.

The intended prediction time is order placement. Only information available at that point should enter a deployed model. The original idea includes price, discount, customer history and shipping details. The release provides product-level averages and shipping country, not exact order prices, shipping methods or delivery dates. These distinctions guide feature selection.

Why machine learning?

A fixed price-threshold rule cannot flexibly represent interactions between product type, brand and customer context. Supervised learning can estimate associations from labeled examples and produce a score evaluated on unseen data. Whether it improves upon simple rules remains an empirical question: majority-class and simple rule baselines will be compared with learned models in the next phase.

Current deliverable

This submission implements documented acquisition, source verification, data-quality checks, joins, a reproducible sample, train-fitted preprocessing and exploratory analysis. Model training, prediction explanations and online deployment are future milestones. No predictive-performance result is claimed here.

2. Data acquisition and provenance

We selected the **ASOS GraphReturns** release from the original Open Science Framework repository [1], described by McGowan et al. [2]. It contains real, anonymized fashion-retail purchase outcomes, customer nodes and product-variant nodes. The paper describes training purchases from September–October 2021 and a later testing period. This deliverable uses the three *training* files only.

OSF metadata identifies the license as **Creative Commons Attribution 4.0 International**. Attribution, source links and transformations are documented. Raw and processed datasets are excluded from Git; acquisition scripts and evidence are included.

Original training table	Rows	Columns
Purchase events	1,369,133	3
Customers	777,001	30
Product variants	411,495	44

The files total 364,102,152 bytes (about 364 MB). The downloader obtains the original file listing, checks SHA-256 hashes against OSF metadata, and records URLs, sizes and verification timestamps. The loader checks pinned hashes and restricts deserialization to required pandas/numpy object types.

Evidence: The original file listing, dataset/license metadata and timestamped download manifest are saved under `reports/evidence/`.

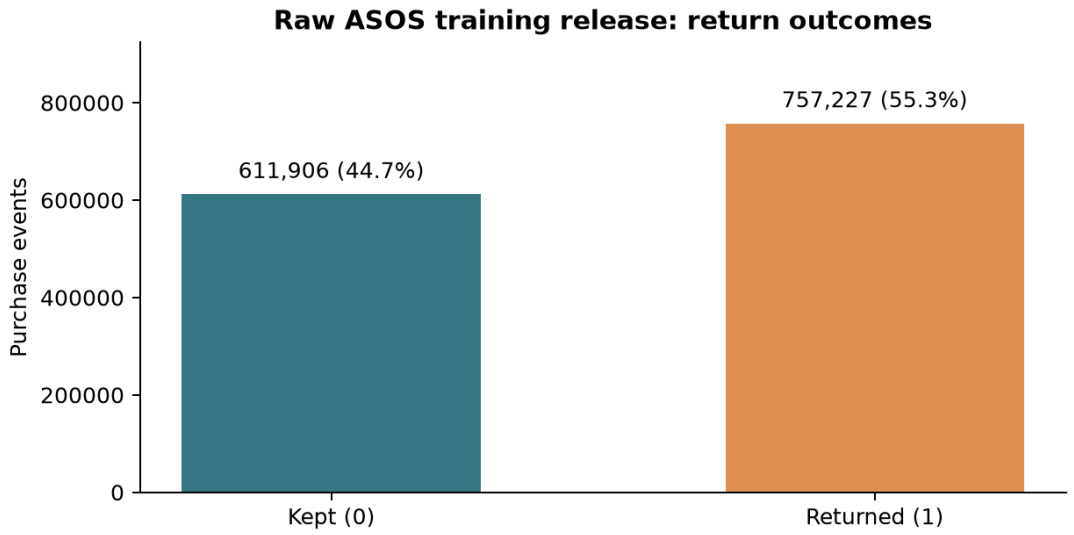


Figure 1. Raw outcomes: 611,906 kept and 757,227 returned events, or approximately 55.31% returned. This is not the retailer's overall return rate.

Representativeness

The paper states that included customers have at least one return, limiting generalization to customers without return history. Most category names are anonymized; `Country_G`, for example, cannot be mapped to a real country. One historical fashion-retail sample does not represent all e-commerce domains. Official testing files remain unused in this phase.

3. Cleaning and data-quality findings

Every raw table is audited before sampling. No missing cells occur in the three files, but relational completeness is a separate issue.

Audit result	Count
Repeated customer–variant–outcome tuples	6,551
Events without a matching customer node	71,648
Events without a matching product node	474,590
Events missing either node (union)	520,679
Events with both nodes available	848,454
Complete events with implausible derived age	12,099
Reproducible development sample	100,000

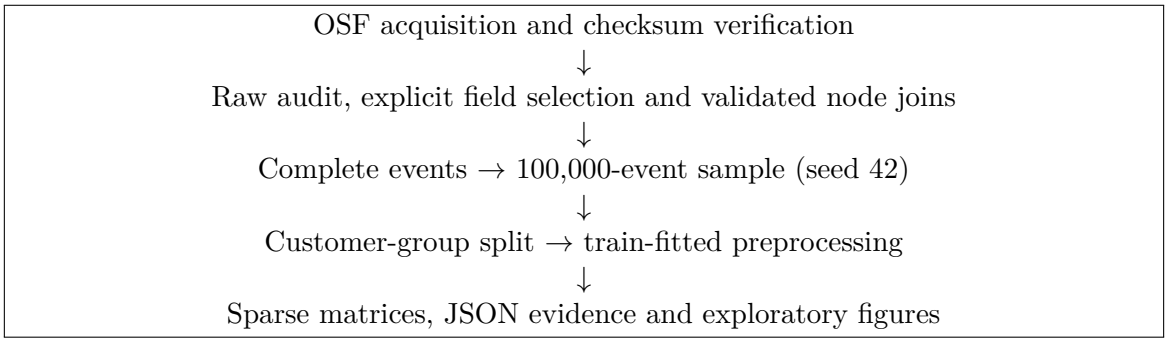
Duplicates. The implementation removes node-table exact duplicates before checking key uniqueness; none were found. Repeated event tuples are retained because no order ID or timestamp distinguishes duplicate records from legitimate repeat purchases. Both node tables repeat a return-reason column name; only explicitly selected, unambiguous fields enter the join.

Joins. Customer and product joins enforce many-to-one relationships. Events missing either node are excluded and counted. Missing-node counts overlap and must not be summed. Approximately 38.03% of events are excluded; their return rate is 56.23%. The exclusion can change the population and needs further investigation.

Missing and invalid values. Negative/nonfinite financial values and zero average prices become missing; none occur among complete events in this run. Approximate age uses the source year, 2021. Ages outside 0–100 become missing for auditing. Age and gender are excluded from predictors. The sampled age column has 1,413 missing values.

Outliers. Valid high prices remain in the cleaned data. Numerical preprocessing caps values at training-sample 1st and 99th percentiles, without learning bounds from validation rows or deleting high-value purchases.

Implemented data flow



The sample is not deliberately class-balanced. Its return fraction is 54.484%.

4. Feature strategy and preprocessing

Field / group	Treatment and rationale
Average product price	Median imputation, percentile clipping and standard scaling. Product-level GBP average, not exact order price.
Average discount	Same pipeline; retained in source units, not assumed to be a percentage or a GBP amount.
Product type, brand, shipping country	Most-frequent imputation and one-hot encoding; unknown categories ignored at transformation time.
Premier membership	Categorical membership indicator.
Customer and variant IDs	Join/split keys only, excluded from features.
Age and gender	Data-quality auditing only.
Sales/returns counts, return rates and reasons	Excluded because their timing relative to each target event is unverified.

Standard scaling uses $z = (x - \mu_{\text{train}})/\sigma_{\text{train}}$. Separate min–max normalization is unnecessary. One-hot encoding avoids an artificial category order.

A seeded customer-group split produces **80,000 training rows** and **20,000 validation rows**, with 75,588 and 18,897 distinct customers and **zero overlap**. All imputation, clipping, scaling and category fitting use training rows only. The sparse matrices contain 35 features.

Training/validation return rates are 54.54%/54.25%. Clipping bounds are GBP 4.16–100.32 for price and 12.86–21.27 for the source discount measure.

Temporal limitation. Customer grouping does not establish chronological validity. Product-average price and discount may include target-period information. Their timing must be verified or the fields replaced before production training. This is a preprocessing demonstration, not a leakage-free deployment benchmark.



Figure 2. Price correlates modestly with return outcome ($r = 0.127$); this is descriptive and does not establish causation.

5. Initial exploratory analysis

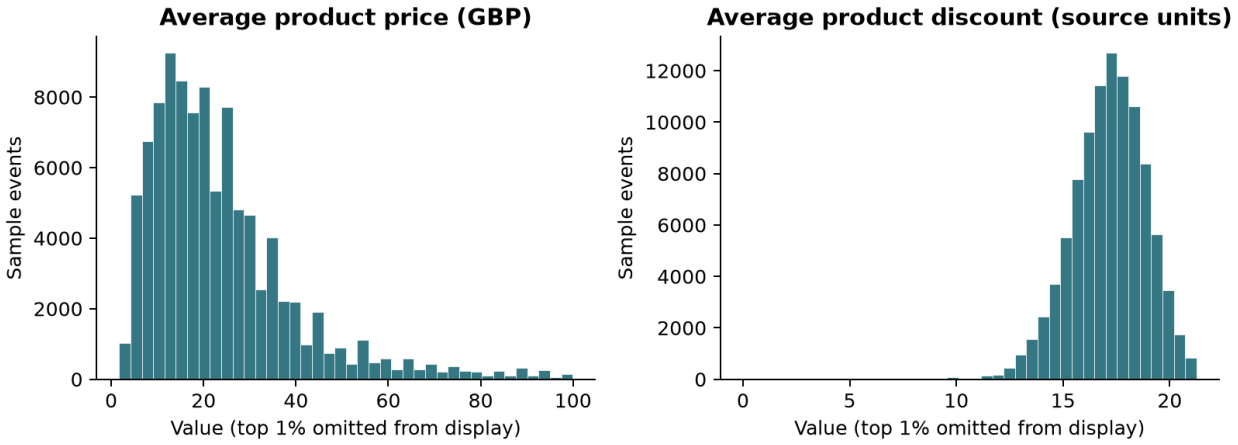


Figure 3. Product-level averages across sampled events. The highest 1% is omitted only from this visualization; those values remain in the cleaned sample.

Average price has a median of GBP 19.70, mean GBP 24.74 and maximum GBP 420.00. The larger mean is consistent with a right tail. Discount has a median of 17.26 in source units. These are event-weighted summaries of product averages, not order-level amounts.

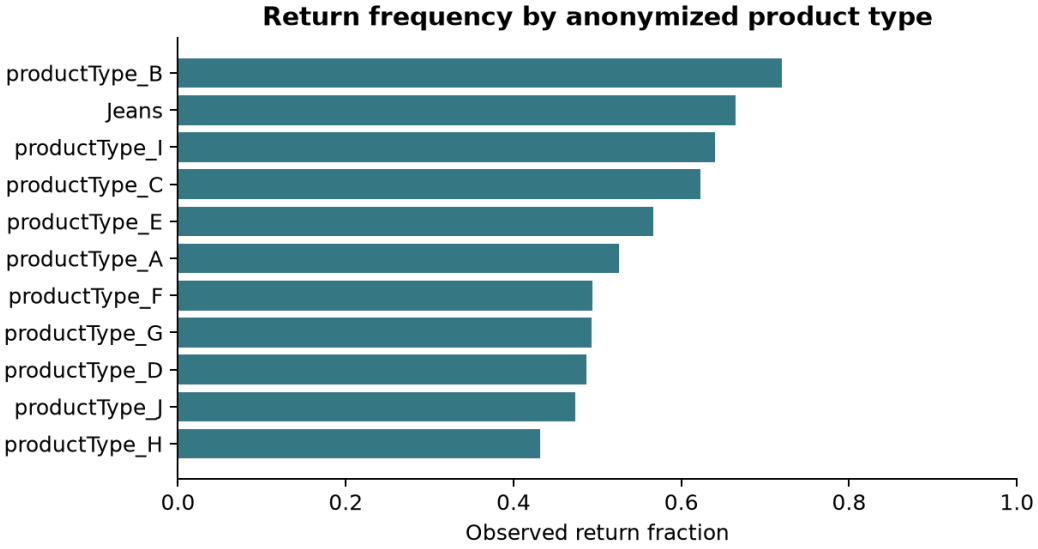


Figure 4. Return frequency ranges from 43.18% for `productType_H` (n = 7,819) to 72.01% for `productType_B` (n = 16,294). The largest group, `productType_J`, has 46,016 events and a 47.35% return rate.

These differences motivate category-aware models, but are not evidence of predictive performance. Product mix, selection and missing-node exclusions may contribute. The weak discount correlation ($r = -0.029$) does not establish that it is useless; interactions may exist.

6. Bias checks, limitations and next phase

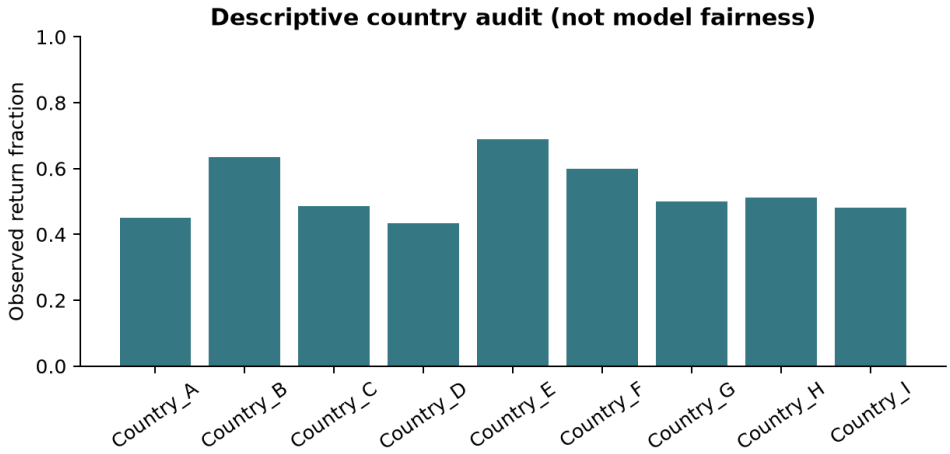


Figure 5. Country return fractions range from 43.43% to 68.79%. Country_G accounts for 55,392 sample events, while several groups have fewer than 2,000.

This is a dataset-level check, not proof of model fairness. Later evaluation should compare precision, recall, false-positive rates and calibration across sufficiently supported groups. Omitting demographics does not remove all bias because other features may act as proxies.

Limitations. Selection on return history, missing node matches, anonymized labels, uncertain aggregate timing and absent order baskets/dates constrain the work. No official test-period evaluation, calibrated probability or business-savings claim is made.

Planned Track A implementation

1. Resolve aggregate timing and missing-node provenance; assess a later-period test.
2. Compare majority/rule baselines, logistic regression and a tree classifier. Report precision, recall, F1, PR-AUC, ROC-AUC, confusion matrices and calibration.
3. Choose thresholds on validation data using explicit false-alarm/missed-return costs.
4. Host the interface and backend, reusing the fitted preprocessing pipeline.

Team workflow

Proposed responsibilities: Soban handles problem definition, report integration and coordination; Mahad handles acquisition, provenance and cleaning; Taimur handles preprocessing, EDA and validation. Each reviews and runs their part and commits under their own account. This is a work allocation, not a claim that all reviews and commits have occurred.

Generative AI assisted with drafting and implementation. Members remain responsible for understanding and validating the work. The README gives the execution sequence; JSON evidence and chart scripts make results inspectable.

References

[1] ASOS. *ASOS graph returns dataset*. OSF, 2022. https://osf.io/c793h/. CC BY 4.0. Accessed 24 October 2026.

[2] J. McGowan et al. *A Dataset for Learning Graph Representations to Predict Customer Returns in Fashion Retail*. 2023. https://arxiv.org/abs/2302.14096.